STREAMFLIX DATA ANALYTICS PROJECT

Phase 1: Data Cleaning & Quality Control Report

Dataset Preparation and Validation

Project	StreamFlix
Phase	Phase 1 — Data Cleaning & Quality Control
Workbook	streamflix.xlsx
Tables	6 datasets / sheets
Prepared for	Data Analytics Project

Purpose: document the structure, cleaning activities and quality-control checks performed to prepare the StreamFlix data for exploratory analysis, SQL KPI calculation and Power BI reporting.

1. Project Overview

The objective of Phase 1 was to prepare the StreamFlix datasets for reliable analysis. The work focused on reviewing data quality, standardising fields, checking data types and organising the cleaned datasets into a single Excel workbook. This phase provides the data foundation for the later EDA, SQL and Power BI phases.

2. Dataset Overview

The project contains six related datasets. The table below records the initial row counts supplied in the project brief/work notes; these are dataset sizes, not a claim that every row passed every validation check.

Dataset / Sheet	Initial Records	Purpose
subscribers	15,000	Subscriber profile and subscription information
titles	9,000	Content catalogue and title attributes
watch_history	650,000	Viewing sessions and watch activity
ratings	130,000	Subscriber ratings for content
reviews	110,000	Written reviews and sentiment information
watchlist	65,000	Titles saved by subscribers
Total	979,000	Across all six datasets

3. Data Cleaning Activities

Missing-value review: Blank and null values were reviewed to identify fields that may affect analysis. Missing values should be handled according to field meaning; for example, an empty churn date may be valid for an active subscriber and should not automatically be filled.

Duplicate review: Exact duplicate rows and potential duplicate records in key identifier columns were included in the quality-control review. Duplicate records should only be removed after checking the relevant business key and confirming that the repeated row is not a valid event.

Data-type review: Identifier fields, dates, numeric measures and category fields were reviewed for appropriate data types. Dates should be stored as dates, watch duration and fees as numeric values, and category fields as consistent text.

Text standardisation: Text fields were reviewed for avoidable leading/trailing spaces and inconsistent category labels. Standardised labels help prevent the same category from appearing as multiple values in reports.

Invalid-value and date review: Numeric and date fields were considered for invalid or implausible values, such as negative watch duration, invalid ratings, or dates that do not make business sense. Any exceptions should be recorded with the correction or reason for retention.

4. Relationship and Business-Rule Checks

The datasets are related through subscriber and title identifiers. The following relationships are important to validate before combining tables for analysis:

• subscribers to watch_history — subscriber identifiers should match valid subscriber records.

• titles to watch_history — title identifiers should match valid catalogue records.

• subscribers and titles to ratings, reviews and watchlist — referenced subscriber and title identifiers should exist in their parent datasets.

• Business-rule checks — values such as rating scores, watch duration, subscription dates and status fields should follow the definitions in the project brief.

5. Final Workbook Organisation

The cleaned datasets were organised into one Excel workbook named **streamflix.xlsx**, with separate sheets for **subscribers**, **titles**, **watch_history**, **ratings**, **reviews** and **watchlist**. Keeping the tables separate preserves their structure and makes them easier to import into Python, SQL Server and Power BI.

6. Quality-Control Summary and Evidence Status

The workbook structure and six expected datasets have been organised, with 979,000 initial records in total according to the supplied project counts. The cleaning and validation areas listed in this report describe the checks to document. Exact measured counts for missing values, duplicate rows, invalid records and unmatched foreign keys were not available in the information supplied for preparing this report; therefore, this report does not invent those figures or state that all checks passed.

Quality Check	Evidence to record in final QC log
Missing values	Null count and percentage for important columns; action taken
Duplicates	Exact duplicate count and duplicate primary/business-key count
Data types	Columns converted or confirmed; conversion errors, if any
Invalid values	Count of invalid values and correction/exclusion decision
Key relationships	Unmatched subscriber_id/title_id counts for each related table

7. Tools Used

Microsoft Excel was used to organise and prepare the six datasets. The workbook structure supports subsequent analysis in Python, SQL Server and Power BI.

8. Conclusion

Phase 1 established the cleaned-data workbook structure required for the StreamFlix analytics project. Before marking the quality-control stage as fully verified, the final QC log should include the actual counts and outcomes for missing values, duplicates, invalid records and key-relationship checks. Once these evidence fields are completed, the dataset is ready to serve as the input for Phase 2 EDA, Phase 3 SQL analysis and Phase 4 Power BI dashboard development.

Note: This report is based on the project details and dataset counts provided in the conversation. Replace or supplement the evidence-status section with actual QC results from the final workbook if those checks have been measured separately.